

IdentityManager

-identifier: PatientIdentifier
-mismatchLog: List<String>
-anomalyThreshold

+processIncomingData(int patientId, double measurementValue, String recordType, long timestamp): HospitalPatient
+verifyIdentity(int patientId): boolean
+handleMismatch(int patientId): void
+logAnomaly(String message): void

+uses

PatientIdentifier

-dataStorage: DataStorage
-toleranceLevel: int

+queries

+matchPatient(int patientId): HospitalPatient
+validateIdentifier(int patientId): boolean
+retrievePatientInfo(int patientId): HospitalPatient

DataStorage

-patientMap: Map<Integer, HospitalPatient>
+addPatientData(int patientId, double measurementValue, String recordType, long timestamp): void
+getRecords(int patientId, long startTime, long endTime): List<PatientRecord>
+getPatient(int patientId): HospitalPatient
+getAllPatients(): List<HospitalPatient>

PatientRecord

-patientId: int
-recordType: String
-measurementValue: double
-timestamp: long

+PatientRecord(int patientId, double measurementValue, String recordType, long timestamp)
+getPatientId(): int
+getMeasurementValue(): double
+getTimestamp(): long
+getRecordType(): String

HospitalPatient

-patientId: int
-name: String
-medicalHistory: List<PatientRecord>

+contains

+getPatientId(): int
+getName(): String
+getMedicalHistory(): List<PatientRecord>
+addRecord(record PatientRecord): void